

ABDUL RAZAK

Staff Data Scientist, Extreme Networks

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Portfolio | [Google Scholar](#) | [DBLP](#) | [ORCID](#) | [Patents](#) | [LinkedIn](#)

RESEARCH SUMMARY

My work concerns decision making when objectives conflict and preferences are known only in part. During my doctorate and post-doctoral fellowship I developed exact and approximate algorithms for multi-objective constraint optimisation and influence diagrams, with methods that elicit only as much preference information as a decision requires. That work appeared at AAI, IJCAI, UAI, CP and ICTAI, largely with colleagues at IBM Research Dublin. In industry I applied the same tools to building security, where the work produced eight granted patents, and since 2020 to enterprise wireless networks, where I lead the data-science work on generative models for network diagnosis.

Research interests. Multi-objective decision making under uncertainty; preference elicitation and inference; probabilistic graphical models; grounded and retrieval-augmented generation; machine learning for networked systems.

EDUCATION

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| 2010–2014 | PhD, Computer Science (Artificial Intelligence)
National University of Ireland, Cork . Cork Constraint Computation Centre
Thesis: <i>Reasoning with Imprecise Trade-offs in Decision Making under Certainty and Uncertainty</i> , supervised by Nic Wilson. Funded by the IRCSET / IBM Enterprise Partnership Scheme. |
| 2003–2005 | MSc, Mathematics with Computer Science
Osmania University, Hyderabad, India
Awarded the university merit scholarship. |
| 2000–2003 | BSc, Mathematics, Statistics and Computer Science
Osmania University, Hyderabad, India |

PROFESSIONAL APPOINTMENTS

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| Since 2020 | Staff Data Scientist
Extreme Networks . Scottsdale, Arizona ; previously Cork, Ireland <ul style="list-style-type: none">– Lead the data-science work on generative models for network diagnosis. Language models generate and explain hypotheses; anomaly detection, a knowledge graph and live telemetry supply the evidence they reason over.– Built agent workflows that triage client connectivity, roaming and device-health problems across thousands of access points and switches, and draft remediation steps for network operators.– Developed correlation and inference pipelines over telemetry and client-experience metrics, so that investigation of routine faults takes seconds rather than hours.– Adapted the preference-inference methods from my research to rank recommendations for customers whose priorities differ in reliability, cost and coverage. |
| 2017–2020 | Data Scientist
Johnson Controls . Cork, Ireland <ul style="list-style-type: none">– Machine-learning lead for the intrusion-alarm, access-control and retail-analytics product lines.– Alarm-panel classification by Pareto optimisation, root-cause analysis of events, and site-independent signature generation. Eight granted US patents issued from this work.– Relational probabilistic models in BLOG and Bayesian inference in PyMC3, deployed across large multi-site estates. |
| 2014–2017 | Post-Doctoral Fellow, Science Foundation Ireland
Insight Centre for Data Analytics , NUI Cork <ul style="list-style-type: none">– Probabilistic graphical models, clustering and influence diagrams for industrial decision problems, within a programme run with United Technology Research Centre.– Preference inference with lexicographic and weighted-average models, partly with IBM Research Dublin. |
| 2005–2010 | Lecturer in Mathematics
Osmania University . Hyderabad, India <ul style="list-style-type: none">– Undergraduate teaching in algebra, real and numerical analysis, calculus, probability and statistics. |

PEER-REVIEWED PUBLICATIONS

Listed most recent first. Complete record on [Google Scholar](#) and [DBLP](#).

- 2017 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with possibly optimal policies. *Proceedings of the 31st AAAI Conference on Artificial Intelligence*.
- 2015 Anne-Marie, **A. Razak**, N. Wilson. The comparison of multi-objective preference inference based on lexicographic and weighted average models. *Proceedings of the 27th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*.
- 2015 N. Wilson, **A. Razak**, R. Marinescu. Computing possibly optimal solutions for multi-objective constraint optimisation with trade-offs. *Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI)*.
- 2015 Anne-Marie, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with lexicographic preference models. *Workshop on Algorithmic Issues for Inference in Graphical Models (AIGM)*.
- 2013 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with trade-offs. *Proceedings of the 19th International Conference on Principles and Practice of Constraint Programming (CP)*.
- 2012 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with trade-offs. *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence (UAI)*.

GRANTED US PATENTS

Nine granted between 2019 and 2021, grouped by subject. Filings are searchable on [Justia](#) and [Google Patents](#).

Decision intelligence and maintenance

- 2019 **A. Razak**, L. A. Deleris, R. Marinescu, N. Wilson. Interactive method to reduce the amount of trade-off information required from decision-maker in multi-attribute decision-making under uncertainty.
- 2019 **A. Razak**, G. Subramanian, C. J. Donovan. Virtual maintenance manager.
- 2021 D. Horgan, **A. Razak**, G. Bernardi. Root-cause analysis of event occurrences.

Alarm management

- 2020 **A. Razak**, M. Stewart, G. Subramanian. Automated alarm panel classification using Pareto optimization.
- 2021 **A. Razak**. Systems and methods for managing alarm data of multiple locations.
- 2021 M. Costello, F. Fala, J. R. Holliday, N. B. Pavel, **A. Razak**. Building security system with alarm lists and alarm rule editing.

Building security analytics

- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Building security analysis system with site-independent signature generation for predictive security analysis.
- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Rules-based method of identifying misuse of emergency fire exits using data generated by a security alarm system.
- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Systems and methods for managing a security system.

HONOURS AND FUNDING

- IEEE Senior Member, elevated May 2026.
- Science Foundation Ireland post-doctoral fellowship, Insight Centre for Data Analytics, NUI Cork, 2014 to 2017.
- IRCSET / IBM Enterprise Partnership Scheme scholarship, full doctoral funding, 2010 to 2013.
- Osmania University merit scholarship (MSc, 2003 to 2005), and 40th place of roughly 30,000 candidates in the university mathematics entrance examination.

ACADEMIC SERVICE AND INVITED TALKS

- Reviewer, *Computers and Electrical Engineering* (Elsevier). More than ten manuscripts in artificial intelligence, wireless networking and intelligent systems.
- Reviewer, IEEE International Conference on Communications (ICC), artificial-intelligence submissions to the 2025 technical programme.
- Invited talks: Sapienza University of Rome, on preference handling in decision making under certainty and uncertainty; NUI Galway, on probabilistic decision making under uncertainty; NUI Cork and IBM Research Dublin, during the Risk Management Collaboratory and United Technology Research Centre programmes.
- Conference presentations, oral and poster, at AAAI, IJCAI, UAI, CP and ICTAI.
- Five years of undergraduate lecturing, and mentoring of junior data scientists in industry.

TECHNICAL SKILLS

Programming	Python, C++ (STL, Boost, Eigen), SQL, PySpark, PyFlink.
Modelling	PyMC3, BLOG, Swift, ILOG CPLEX, LPSOLVE, scikit-learn.
Platforms	AWS SageMaker and Athena, Azure Databricks, streaming and batch pipelines.
Languages	English, Hindi, Telugu, Urdu. Irish citizen since 2021.